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United States Patent	12399278
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Yavid; Dmitriy

Hybrid LIDAR with optically enhanced scanned laser

Abstract

An autonomous vehicle LIDAR system includes: a laser; optical transmission system to shape/scan the light beam along light paths toward a target; an optical reception system collecting reflected laser light; and an electronic system synchronizing the beam scan, performing time-of-arrival measurements, and determining target range. The reception system includes an objective lens and a one-dimensional sensor array, each sensor formed of a grid of sub-pixels. A first transmission system includes: a cylindrical lens, scanning mirror, and collimating lens. The cylindrical lens expands the laser beam only in a first orthogonal direction to the optical path, producing an oval beam. A second transmission system includes: a focusing lens, scanning mirror, and second lens with lenslets being sequentially illuminated by the optical paths that individually focus the light that's collimated by a third lens creating spots on the target. Each sensor includes an FET that selectively couples sub-pixels to a readout channel.

Inventors:	**Yavid; Dmitriy (Stony Brook, NY)**
Applicant:	**Red Creamery LLC** (Massapequa Park, NY)
Family ID:	**1000005767426**
Assignee:	**Red Creamery LLC (Massapequa Park, NY)**
Appl. No.:	**17/356676**
Filed:	**June 24, 2021**

Related U.S. Application Data

continuation parent-doc US 15432105 20170214 US 10571574 20200225 child-doc US 16744410
continuation-in-part parent-doc US 17000464 20200824 US 11556000 child-doc US 17356676
continuation-in-part parent-doc US 16744410 20200116 US 11156716 child-doc US 17000464
us-provisional-application US 63154990 20210301
us-provisional-application US 62890189 20190822
us-provisional-application US 62295210 20160215

Publication Classification

Int. Cl.: **G01C3/08** (20060101); **G01S7/481** (20060101); **G01S7/4863** (20200101); **G01S7/4865** (20200101); **G01S17/89** (20200101)

U.S. Cl.:

CPC **G01S17/89** (20130101); **G01S7/4817** (20130101); **G01S7/4863** (20130101); **G01S7/4865** (20130101);

Field of Classification Search

CPC: G01S (17/89); G01S (7/4817); G01S (7/4863); G01S (7/4865)

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9407285	12/2015	Kozak	N/A	N/A
9420177	12/2015	Pettegrew	N/A	N/A
9420264	12/2015	Gilliland	N/A	N/A
9425654	12/2015	Lenius	N/A	N/A
9448110	12/2015	Wong	N/A	N/A
9453907	12/2015	Zheleznyak	N/A	N/A
9453914	12/2015	Stettner	N/A	N/A
9453941	12/2015	Stainvas Olshansky	N/A	N/A
9465112	12/2015	Stettner	N/A	N/A
9470520	12/2015	Schwarz	N/A	N/A
9476968	12/2015	Anderson	N/A	N/A
9476983	12/2015	Zeng	N/A	N/A
9489746	12/2015	Sebastian	N/A	N/A
9495466	12/2015	Geringer	N/A	N/A
9519979	12/2015	Hilde	N/A	N/A
9523772	12/2015	Rogan	N/A	N/A
9525863	12/2015	Nawasra	N/A	N/A
9529087	12/2015	Stainvas Olshansky	N/A	N/A
9530062	12/2015	Nguyen	N/A	N/A
9547074	12/2016	Schulz	N/A	N/A
9575162	12/2016	Owechko	N/A	N/A
9575164	12/2016	Kim	N/A	N/A
9575184	12/2016	Gilliand	N/A	N/A
9575341	12/2016	Heck	N/A	N/A

9588220	12/2016	Rondeau	N/A	N/A
9599468	12/2016	Walsh	N/A	N/A
9599714	12/2016	Imaki	N/A	N/A
9602224	12/2016	McLaughlin	N/A	N/A
9606236	12/2016	Rojas	N/A	N/A
9625580	12/2016	Kotelnikov	N/A	N/A
9625582	12/2016	Gruver	N/A	N/A
9651658	12/2016	Pennecot	N/A	N/A
9658322	12/2016	Lewis	N/A	N/A
9658337	12/2016	Ray	N/A	N/A
9678199	12/2016	Hutson	N/A	N/A
9702975	12/2016	Brinkmeyer	N/A	N/A
9710714	12/2016	Chen	N/A	N/A
9735885	12/2016	Sayyah	N/A	N/A
9753124	12/2016	Hayes	N/A	N/A
9753462	12/2016	Gilliland	N/A	N/A
9759809	12/2016	Derenick	N/A	N/A
9772399	12/2016	Schwarz	N/A	N/A
9778362	12/2016	Rondeau	N/A	N/A
9784840	12/2016	Pedersen	N/A	N/A
9790924	12/2016	Bayon	N/A	N/A
9791555	12/2016	Zhu	N/A	N/A
9791557	12/2016	Wyrwas	N/A	N/A
9797995	12/2016	Gilliland	N/A	N/A
9804264	12/2016	Villeneuve	N/A	N/A
9810775	12/2016	Welford	N/A	N/A
9810776	12/2016	Sapir	N/A	N/A
9810777	12/2016	Williams	N/A	N/A
9810786	12/2016	Welford	N/A	N/A
9812838	12/2016	Villeneuve	N/A	N/A
9823118	12/2016	Doylend	N/A	N/A
9823350	12/2016	Fluckiger	N/A	N/A
9823351	12/2016	Haslim	N/A	N/A
9823353	12/2016	Eichenholz	N/A	N/A
9830509	12/2016	Zang	N/A	N/A
9831630	12/2016	Lipson	N/A	N/A
9834209	12/2016	Stettner	N/A	N/A
9841495	12/2016	Campbell	N/A	N/A
9851433	12/2016	Sebastian	N/A	N/A
9851442	12/2016	Lo	N/A	N/A
RE46672	12/2017	Hall	N/A	N/A
9857473	12/2017	Kim	N/A	N/A
9860770	12/2017	McLaughlin	N/A	N/A
9869753	12/2017	Eldada	N/A	N/A
9869754	12/2017	Campbell	N/A	N/A
9870512	12/2017	Rogan	N/A	N/A
9872010	12/2017	Tran	N/A	N/A
9874635	12/2017	Eichenholz	N/A	N/A
9877009	12/2017	Tran	N/A	N/A
9880263	12/2017	Droz	N/A	N/A

9880281	12/2017	Gilliland	N/A	N/A
9881220	12/2017	Korvadi	N/A	N/A
9882433	12/2017	Lenius	N/A	N/A
9885778	12/2017	Dussan	N/A	N/A
9891711	12/2017	Lee	N/A	N/A
9892567	12/2017	Binion	N/A	N/A
9897687	12/2017	Campbell	N/A	N/A
9897689	12/2017	Dussan	N/A	N/A
9904375	12/2017	Donnelly	N/A	N/A
9905032	12/2017	Rogan	N/A	N/A
9905987	12/2017	Seo	N/A	N/A
9905992	12/2017	Welford	N/A	N/A
9910136	12/2017	Heo	N/A	N/A
9910139	12/2017	Pennecot	N/A	N/A
9910155	12/2017	Lundquist	N/A	N/A
9915726	12/2017	Bailey	N/A	N/A
9921297	12/2017	Jungwirth	N/A	N/A
9921307	12/2017	Schmalengurg	N/A	N/A
9927524	12/2017	Kaiser	N/A	N/A
9933513	12/2017	Dussan	N/A	N/A
9933514	12/2017	Gylys	N/A	N/A
9945950	12/2017	Newman	N/A	N/A
9958545	12/2017	Eichenholz	N/A	N/A
9971024	12/2017	Schwarz	N/A	N/A
9971035	12/2017	Imaki	N/A	N/A
9983297	12/2017	Hall	N/A	N/A
9983590	12/2017	Templeton	N/A	N/A
9985071	12/2017	Irish	N/A	N/A
9989969	12/2017	Eustice	N/A	N/A
10000000	12/2017	Marron	N/A	N/A
10012474	12/2017	Teetzel	N/A	N/A
10012723	12/2017	Lindskog	N/A	N/A
10012732	12/2017	Eichenholz	N/A	N/A
10018711	12/2017	Sebastian	N/A	N/A
10018725	12/2017	Liu	N/A	N/A
10018726	12/2017	Hall	N/A	N/A
10019803	12/2017	Venable	N/A	N/A
10024964	12/2017	Pierce	N/A	N/A
10031214	12/2017	Rosenweig	N/A	N/A
10031231	12/2017	Zermas	N/A	N/A
10031232	12/2017	Zohar	N/A	N/A
10032369	12/2017	Korvadi	N/A	N/A
10036801	12/2017	Reterrath	N/A	N/A
10036803	12/2017	Pacala	N/A	N/A
D826746	12/2017	Qiu	N/A	N/A
10042042	12/2017	Miremadi	N/A	N/A
10042043	12/2017	Dussan	N/A	N/A
10042159	12/2017	Dussan	N/A	N/A
10046187	12/2017	Doten	N/A	N/A
10048359	12/2017	Zhelenyzak	N/A	N/A

10048374	12/2017	Hall	N/A	N/A
10054841	12/2017	Nomura	N/A	N/A
10061019	12/2017	Campbell	N/A	N/A
10061020	12/2017	Slobodyanyuk	N/A	N/A
10061266	12/2017	Christmas	N/A	N/A
10067230	12/2017	Smits	N/A	N/A
10073166	12/2017	Dussan	N/A	N/A
10078133	12/2017	Dussan	N/A	N/A
10078137	12/2017	Ludwig	N/A	N/A
10088557	12/2017	Yeun	N/A	N/A
10088558	12/2017	Dussan	N/A	N/A
10094657	12/2017	Kiss	N/A	N/A
10094916	12/2017	Droz	N/A	N/A
10094925	12/2017	LaChapelle	N/A	N/A
10094928	12/2017	Josset	N/A	N/A
10107914	12/2017	Kalscheur	N/A	N/A
10107915	12/2017	Rozenzweig	N/A	N/A
10109208	12/2017	Cherepinsky	N/A	N/A
10114109	12/2017	Gazit	N/A	N/A
10114112	12/2017	Slobodyanyuk	N/A	N/A
10115024	12/2017	Stein	N/A	N/A
10120076	12/2017	Scheim	N/A	N/A
10121813	12/2017	Eichenholz	N/A	N/A
10126411	12/2017	Gilliland	N/A	N/A
10126412	12/2017	Eldada	N/A	N/A
10131446	12/2017	Stambler	N/A	N/A
10132928	12/2017	Eldada	N/A	N/A
10139478	12/2017	Gaalema	N/A	N/A
10142538	12/2017	Hurd	N/A	N/A
10145941	12/2017	Lee	N/A	N/A
10145944	12/2017	Shchemelinin	N/A	N/A
10145945	12/2017	Harada	N/A	N/A
10148060	12/2017	Hong	N/A	N/A
10151836	12/2017	O'Keeffe	N/A	N/A
10168423	12/2018	Lombrozo	N/A	N/A
10168429	12/2018	Maleki	N/A	N/A
10175344	12/2018	Jungwirth	N/A	N/A
10175361	12/2018	Haines	N/A	N/A
10180493	12/2018	Eldada	N/A	N/A
10185027	12/2018	O'Keeffe	N/A	N/A
10185028	12/2018	Dussan	N/A	N/A
10185033	12/2018	Justice	N/A	N/A
10191156	12/2018	Steinberg	N/A	N/A
10197669	12/2018	Hall	N/A	N/A
10197676	12/2018	Slobodyyanyuk	N/A	N/A
10197765	12/2018	Schulz	N/A	N/A
10203399	12/2018	Retterath	N/A	N/A
10203401	12/2018	Sebastian	N/A	N/A
10209349	12/2018	Dussan	N/A	N/A
10209359	12/2018	Russell	N/A	N/A

10209709	12/2018	Peters	N/A	N/A
10214299	12/2018	Jackowski	N/A	N/A
10215846	12/2018	Carothers	N/A	N/A
10215847	12/2018	Scheim	N/A	N/A
10215848	12/2018	Dussan	N/A	N/A
10215859	12/2018	Steinberg	N/A	N/A
10222474	12/2018	Raring	N/A	N/A
10222477	12/2018	Keilaf	N/A	N/A
10223806	12/2018	Luo	N/A	N/A
10223807	12/2018	Luo	N/A	N/A
10241196	12/2018	Bailey	N/A	N/A
10241198	12/2018	LaChappelle	N/A	N/A
10247811	12/2018	Clifton	N/A	N/A
10254402	12/2018	Lane	N/A	N/A
10254405	12/2018	Campbell	N/A	N/A
10261006	12/2018	Ray	N/A	N/A
10261187	12/2018	Halmos	N/A	N/A
10262234	12/2018	Li	N/A	N/A
10267898	12/2018	Campbell	N/A	N/A
10267918	12/2018	LaChapelle	N/A	N/A
10274377	12/2018	Rabb	N/A	N/A
10274599	12/2018	Schmalenberg	N/A	N/A
D849573	12/2018	Haban	N/A	N/A
10281254	12/2018	Ginsberg	N/A	N/A
10281322	12/2018	Doyland	N/A	N/A
10281564	12/2018	Low	N/A	N/A
10281581	12/2018	Lipson	N/A	N/A
10281582	12/2018	Elooz	N/A	N/A
10288736	12/2018	Lipson	N/A	N/A
10288737	12/2018	Mooney	N/A	N/A
10295656	12/2018	Li	N/A	N/A
10295660	12/2018	McMichael	N/A	N/A
10295668	12/2018	LaChapelle	N/A	N/A
10295670	12/2018	Stettner	N/A	N/A
10295671	12/2018	Grazit	N/A	N/A
10295672	12/2018	Abari	N/A	N/A
10295673	12/2018	Tucker	N/A	N/A
10302746	12/2018	O'Keeffee	N/A	N/A
10302749	12/2018	Droz	N/A	N/A
D850306	12/2018	Bainter	N/A	N/A
10310058	12/2018	Campbell	N/A	N/A
10310087	12/2018	Laddha	N/A	N/A
10317529	12/2018	Shy	N/A	N/A
10317533	12/2018	Cherepinsky	N/A	N/A
10324170	12/2018	Enberg	N/A	N/A
10324185	12/2018	McWhirter	N/A	N/A
10330777	12/2018	Popovich	N/A	N/A
10330778	12/2018	Kaneda	N/A	N/A
10330780	12/2018	Hall	N/A	N/A
10331956	12/2018	Solar	N/A	N/A

10337996	12/2018	Blagojevic	N/A	N/A
10338201	12/2018	Slobodyyanyuk	N/A	N/A
10338202	12/2018	Mashtare	N/A	N/A
10338220	12/2018	Raring	N/A	N/A
10338224	12/2018	Eken	N/A	N/A
10338225	12/2018	Boehmke	N/A	N/A
10340651	12/2018	Drummer	N/A	N/A
10345446	12/2018	Raring	N/A	N/A
10345447	12/2018	Hicks	N/A	N/A
10346695	12/2018	Clifford	N/A	N/A
10351103	12/2018	Yeo	N/A	N/A
10353057	12/2018	Suzuki	N/A	N/A
10353074	12/2018	Justice	N/A	N/A
10353075	12/2018	Buskila	N/A	N/A
10359507	12/2018	Berger	N/A	N/A
10366282	12/2018	Lee	N/A	N/A
10372138	12/2018	Gilliland	N/A	N/A
10377373	12/2018	Stettner	N/A	N/A
10379135	12/2018	Maryfield	N/A	N/A
10379205	12/2018	Dussan	N/A	N/A
10379220	12/2018	Smits	N/A	N/A
10379540	12/2018	Droz	N/A	N/A
10386464	12/2018	Dussan	N/A	N/A
10386465	12/2018	Hall	N/A	N/A
10386467	12/2018	Dussan	N/A	N/A
10386487	12/2018	Wilton	N/A	N/A
10386488	12/2018	Ridderbusch	N/A	N/A
10393863	12/2018	Sun	N/A	N/A
10393877	12/2018	Hall	N/A	N/A
10394345	12/2018	Donnelly	N/A	N/A
10401480	12/2018	Gaalema	N/A	N/A
10401484	12/2018	Lee	N/A	N/A
10401500	12/2018	Yang	N/A	N/A
10401866	12/2018	Rust	N/A	N/A
10408926	12/2018	Slobodyyabnyuk	N/A	N/A
10408936	12/2018	Van Voorst	N/A	N/A
10408939	12/2018	Kim	N/A	N/A
10408940	12/2018	O'Keeffe	N/A	N/A
10416292	12/2018	de Mersseman	N/A	N/A
10418776	12/2018	Welford	N/A	N/A
10422862	12/2018	Gnecchi	N/A	N/A
10422863	12/2018	Choi	N/A	N/A
10422865	12/2018	Irish	N/A	N/A
10429243	12/2018	Yu	N/A	N/A
10429495	12/2018	Wang	N/A	N/A
10429496	12/2018	Weinberg	N/A	N/A
10429507	12/2018	Sebastian	N/A	N/A
10429511	12/2018	Bosetti	N/A	N/A
10430970	12/2018	Bier	N/A	N/A
10436882	12/2018	Meng	N/A	N/A

10436904	12/2018	Moss	N/A	N/A
10436907	12/2018	Murray	N/A	N/A
10444330	12/2018	Stann	N/A	N/A
10444356	12/2018	Wu	N/A	N/A
10444362	12/2018	Schaefer	N/A	N/A
10444367	12/2018	Lodden	N/A	N/A
10445928	12/2018	Nehmadi	N/A	N/A
10447973	12/2018	Droz	N/A	N/A
10451716	12/2018	Hughes	N/A	N/A
10451740	12/2018	Pei	N/A	N/A
10451742	12/2018	Christmas	N/A	N/A
10458904	12/2018	Batholomew	N/A	N/A
10466342	12/2018	Zhu	N/A	N/A
10469753	12/2018	Yang	N/A	N/A
10473763	12/2018	Schwarz	N/A	N/A
10473767	12/2018	Xiang	N/A	N/A
10473768	12/2018	Walsh	N/A	N/A
10473770	12/2018	Zhu	N/A	N/A
10473784	12/2018	Puglia	N/A	N/A
10474160	12/2018	Huang	N/A	N/A
10474161	12/2018	Huang	N/A	N/A
10481267	12/2018	Wang	N/A	N/A
10481268	12/2018	Vlaiko	N/A	N/A
10482740	12/2018	Fang	N/A	N/A
10488495	12/2018	Sebastian	N/A	N/A
10488496	12/2018	Campbell	N/A	N/A
10488497	12/2018	Cheong	N/A	N/A
10491052	12/2018	Lenius	N/A	N/A
10491855	12/2018	Gates	N/A	N/A
10495757	12/2018	Dussan	N/A	N/A
10502813	12/2018	Schultz	N/A	N/A
10503174	12/2018	Lim	N/A	N/A
10503175	12/2018	Agarwal	N/A	N/A
10509111	12/2018	Park	N/A	N/A
10509112	12/2018	Pan	N/A	N/A
10509120	12/2018	Bilik	N/A	N/A
10509198	12/2018	Zhou	N/A	N/A
10514444	12/2018	Donovan	N/A	N/A
10514447	12/2018	Schwarz	N/A	N/A
10520591	12/2018	Kotelnikov	N/A	N/A
10520592	12/2018	Droz	N/A	N/A
10520602	12/2018	Villeneuve	N/A	N/A
10523880	12/2018	Gassend	N/A	N/A
10527726	12/2019	Bartlett	N/A	N/A
10531004	12/2019	Wheeler	N/A	N/A
10534074	12/2019	Slobodyyanyuk	N/A	N/A
10534079	12/2019	Kim	N/A	N/A
10539116	12/2019	Davoust	N/A	N/A
10539661	12/2019	Hall	N/A	N/A
10539663	12/2019	Liu	N/A	N/A

10545222	12/2019	Hall	N/A	N/A
10545238	12/2019	Rezk	N/A	N/A
10545240	12/2019	Campbell	N/A	N/A
10545289	12/2019	Chriqui	N/A	N/A
10551501	12/2019	LaChapelle	N/A	N/A
10552691	12/2019	Li	N/A	N/A
10556585	12/2019	Berger	N/A	N/A
10557923	12/2019	Watnik	N/A	N/A
10557924	12/2019	Jang	N/A	N/A
10557926	12/2019	Gilliland	N/A	N/A
10557927	12/2019	Marron	N/A	N/A
10557929	12/2019	Kajiyama	N/A	N/A
10557939	12/2019	Campbell	N/A	N/A
10557940	12/2019	Eichenholz	N/A	N/A
10557942	12/2019	Belsey	N/A	N/A
10564261	12/2019	Huebner	N/A	N/A
10564263	12/2019	Efimov	N/A	N/A
10564266	12/2019	O'Keeffe	N/A	N/A
10564285	12/2019	Belsley	N/A	N/A
10565457	12/2019	Luo	N/A	N/A
10571552	12/2019	Gao	N/A	N/A
10571567	12/2019	Campbell	N/A	N/A
10571570	12/2019	Paulsen	N/A	N/A
10571574	12/2019	Yavid	N/A	N/A
10571683	12/2019	Low	N/A	N/A
10576011	12/2019	Krishnan	N/A	N/A
10578717	12/2019	Bucina	N/A	N/A
10578719	12/2019	O'Keeffe	N/A	N/A
10578720	12/2019	Hughes	N/A	N/A
10578721	12/2019	Jang	N/A	N/A
10578724	12/2019	Droz	N/A	N/A
10578742	12/2019	Guo	N/A	N/A
10585174	12/2019	Gnecchi	N/A	N/A
10585175	12/2019	Reterath	N/A	N/A
10591598	12/2019	Jeong	N/A	N/A
10591599	12/2019	O'Keeffe	N/A	N/A
10591600	12/2019	Villeneuve	N/A	N/A
10591601	12/2019	Hicks	N/A	N/A
10591604	12/2019	Xu	N/A	N/A
10591740	12/2019	McMichael	N/A	N/A
10598769	12/2019	Rodrigo	N/A	N/A
10598770	12/2019	Singer	N/A	N/A
10598788	12/2019	Dussan	N/A	N/A
10598791	12/2019	Jain	N/A	N/A
10598922	12/2019	Low	N/A	N/A
10600930	12/2019	Suzuki	N/A	N/A
10605899	12/2019	Singer	N/A	N/A
10605900	12/2019	Spuler	N/A	N/A
10605918	12/2019	Wong	N/A	N/A
10605924	12/2019	Slutsky	N/A	N/A

RE47942	12/2019	Hall	N/A	N/A
D882430	12/2019	Haban	N/A	N/A
10613200	12/2019	Hallstig	N/A	N/A
10613201	12/2019	Pacala	N/A	N/A
10613204	12/2019	Warke	N/A	N/A
10613224	12/2019	Jeong	N/A	N/A
10620301	12/2019	Wilton	N/A	N/A
10620302	12/2019	Zhu	N/A	N/A
10620315	12/2019	Zellinger	N/A	N/A
10620317	12/2019	Chai	N/A	N/A
10620318	12/2019	Yi	N/A	N/A
10627490	12/2019	Hall	N/A	N/A
10627491	12/2019	Hall	N/A	N/A
10627492	12/2019	Shand	N/A	N/A
10627495	12/2019	Gaalema	N/A	N/A
10627512	12/2019	Hicks	N/A	N/A
10627516	12/2019	Eichenholz	N/A	N/A
10629072	12/2019	Felix	N/A	N/A
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Primary Examiner: Hellner; Mark

Attorney, Agent or Firm: O'Rourke IP Law, PLLC

Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS (1) This application claims priority on U.S. Provisional Patent Application Ser. No. 63/154,990, filed on Mar. 1, 2021; and this application is also a continuation in part of U.S. application Ser. No. 17/000,464, filed on Aug. 24, 2020, which claims priority on U.S. Provisional Application Ser. No. 62/890,189, filed on Aug. 22, 2019; and this application is also a continuation in part of U.S. application Ser. No. 16/744,410, filed on Jan. 16, 2020, which is a continuation of U.S. application Ser. No. 15/432,105, filed on Feb. 14, 2017, now issued as U.S. Pat. No. 10,571,574, which claims priority on U.S. Provisional Patent Application Ser. No. 62/295,210, filed on Feb. 15, 2016, all disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates generally to the field of optical Time-of-Flight (ToF) measurement, and more specifically, to directionally-resolved ToF measurement technology, known as Laser Radar (LADAR).
BACKGROUND OF THE INVENTION
(2) In the modern era, beginning in World War II, radio detection and ranging (RADAR) systems were deployed, and which utilize reflected radio waves to identify the position of enemy aircraft. Sonar similarly uses sound waves to locate vessels within the oceans. Soon after the development of laser technologies, light/laser detection and ranging (LIDAR/LADAR) systems underwent development.
(3) LADAR is generally based on emitting short pulses of light within certain Field-Of-View (FOV) at precisely-controlled moments, collecting the reflected light and determining its Time-of-Arrival, possibly, separately from different directions. Subtraction of the pulse emission time from ToA yields ToF, and that, in turns, allows one to determine the distance to the target the light was reflected from. LADAR is the most promising vision technology for autonomous vehicles of different kinds, as well as surveillance, security, industrial automation, and many other areas, where detailed information about the immediate surroundings is required. While lacking the range of radar, LADAR has a much higher resolution due to much shorter wavelengths that are used for sensing, and hence, comparatively relaxed diffractive limitations. It may be especially useful for moving vehicles, both manned, self-driving, and unmanned, if it could provide detailed 3D information in real-time, with the potential to revolutionize vehicles' sensing abilities and enable a variety of missions and applications.
(4) However, until recently, LADARs have been prohibitively large and expensive for vehicular use. They were also lacking in desirable performance: to become a true real-time vision technology, LADAR must provide high-resolution imagery, at high frame rates, comparable with video cameras, in the range of 15-60 fps, and cover a substantial solid angle. Ideally, a LADAR with omnidirectional coverage of 360° azimuth and 180° elevation would be very beneficial. Collectively, these requirements may be called “real-time 3D vision”.

(5) A variety of approaches has been suggested and tried, including mechanical scanning, non-mechanical scanning, and imaging time-of-flight (ToF) focal-plane arrays (FPA). There is also a variety of laser types, detectors, signal processing techniques, etc. that have been used to date, as shown by the following.
(6) U.S. Patent Application Pub. No. 2012/0170029 by Azzazy teaches 2D focal plane array (FPA) in the form of a micro-channel plate, illuminated in its entirety by a short power pulse of light. This arrangement is generally known as flash LADAR.
(7) U.S. Patent Application Pub. No. 2012/0261516 by Gilliland teaches another embodiment of flash LADAR, with a two dimensional array of avalanche photodiodes illuminated in its entirety as well.
(8) U.S. Patent Application Pub. No. 2007/0035624 by Lubard teaches a similar arrangement with a 1D array of detectors, still illuminated together, while U.S. Pat. No. 6,882,409 to Evans further adds sensitivity to different wavelengths to flash LADAR. Another approach is the use a 2D scanner and only one detector receiving reflected light sequentially from every point in the FOV, as taught by US Patent Application Pub. No. 2012/0236379 by da Silva.
(9) Additional improvements to this approach are offered by U.S. Pat. No. 9,383,753 to Templeton, teaching a synchronous scan of the FOV of a single receiver via an array of synchronized MEMS mirrors. This arrangement is known as retro-reflective.
(10) Yet another approach is to combine multiple lasers and multiple detectors in a single scanned FOV, as taught by U.S. Pat. No. 8,767,190 to Hall.
(11) See also, the laser range camera of U.S. Pat. No. 5,870,180 to Wangler.
(12) The herein disclosed apparatus provides improvements upon certain prior art LADAR systems.
(13) It is noted that the citing of any reference within this disclosure, i.e., any patents, published patent applications, and non-patent literature, is not an admission regarding a determination as to its availability as prior art with respect to the herein disclosed and claimed method/apparatus.
OBJECTS OF THE INVENTION
(14) The present invention is aimed at overcoming the limitations of both scanning and imaging approaches to LADAR by combining their advantages and alleviating their problems, including but not limited to: 1. Improve Signal-to-Noise Ratio (SNR); 2. Lower the peak power of illumination sources, reduce their cost and improve efficiency; 3. Reduce overall cost and power consumption; and 4. Increase the resolution.
(15) Further objects and advantages of the invention will become apparent from the following description and claims, and from the accompanying drawings.
SUMMARY OF THE INVENTION
(16) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.
(17) A laser radar (LIDAR) system for an autonomous vehicle may include: a laser configured to emit a beam of light at a wavelength; an optical transmission system configured to shape the beam of light emitted by the laser, and to scan the beam along a plurality of transmission light paths toward a target; an optical reception system configured to collect the laser light reflected from the target along a plurality of reception light paths; and an electronic control system configured to synchronize the scan of the beam with a respective time-of-arrival measurement, to perform a time-of-arrival measurement, and determine a range of the target.
(18) The optical reception system is formed of an objective lens and a one-dimensional array of light sensing devices ("sensors"), with each sensor of the array being sensitive to the wavelength of light emitted by the laser, and being configured to detect and image the laser light reflected from the target. Each sensor of the one-dimensional sensor array is formed of an n by m grid of sub-

pixels, with each sub-pixel configured to receive the laser light reflected from the target.

(19) A first embodiment of the optical transmission system includes: a first cylindrical lens, a scanning mirror, and a second lens. The first cylindrical lens is positioned and configured to expand the laser beam in a first orthogonal direction to the optical path and to have no effect on the beam in a second orthogonal direction to the optical path, to produce an expanding beam comprising an oval-shaped cross-section. The mirror oscillates about an axis to scan the expanding beam of light along a plurality of optical paths toward the target surface. The second lens is positioned in the plurality of optical paths, and is configured to collimate the scanned expanding beam to produce a collimated oval beam being scanned across the target to create a series of oval spots on the target.

(20) A second embodiment of the optical transmission system includes: a first lens, a scanning mirror, a second lens, and third lens. The first lens is positioned and configured to focus and fit the laser beam onto the mirror. The mirror oscillates about an axis and is positioned to scan the focused light along a plurality of optical paths toward the target surface. The second lens is positioned in the plurality of optical paths, and is formed with a plurality of lenslets, where each of the plurality of lenslets is positioned in line to be sequentially illuminated by the plurality of optical paths. The plurality of lenslets are configured to individually focus the light, which is collimated by the third lens to create a series of spots on the target.

(21) In one embodiment the optical reception system may further include one readout channel per sensor. Each readout channel may include timing circuitry configured to determine the time-of-flight of the beam from the emission from the laser to the reception of the laser light reflected from the target. The optical reception system may further include a field effect transistor (FET). The system may be configured such that only one of the sub-pixels within each sensor is connected to the readout channel through the FET at a time, being any one of the grid of sub-pixels determined by the FET to receive a maximum signal. Alternatively, the system may be configured such that two or more of the sub-pixels within each the sensor is connected to the readout channel through the FET, and the FET is configured to sum output signals of the two or more sub-pixels and transmit the summed output signals to the readout channel. The FET may also be configured to sum output from only the sub-pixels that receive signals.

(22) Hundreds of sensors may be used, and the sensors used may be any suitable sensor known in the art, or which may later be developing, including, but not limited to, avalanche photodiodes. For a LIDAR system to be used in an autonomous vehicle, the laser may advantageously be: a master oscillator power amplifier (MOPA) laser, or a continuous wave (CW) fiber laser. One exemplary embodiment of a LIDAR system for an autonomous vehicle that achieves a 200 meter detection range may utilize: a one-dimensional sensor array formed of 512 sensors; a mirror configured to oscillate at about 80 KHz; and approximately 10 watts of power for the laser. Each sub-pixel may be a rectangle with sides in a range being between five um and ten um, or in a range being between one um and five um.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

(2) The description of the various example embodiments is explained in conjunction with appended drawings, in which:

(3) FIG. **1** illustrates a concept for a 1D scanning and imaging hybrid system.

(4) FIG. **2**A illustrates a 1D hybrid LADAR installed on a Unmanned Aerial Vehicle (UAV) to obtain 2D ToF data.

(5) FIG. **2**B illustrates a 1D hybrid LADAR coupled to an additional scanning mirror to obtain 2D ToF data.
(6) FIG. **2**C illustrates a 1D hybrid LADAR installed on a rotating platform to obtain 2D ToF data.
(7) FIG. **3**A illustrates tight focusing of the laser beam on the scanning mirror of a LADAR and subsequent re-collimation with a cylindrical lens.
(8) FIG. **3**B is a bottom view of the LADAR of FIG. **3**A.
(9) FIG. **3**C is a side view of the LADAR of FIG. **3**A.
(10) FIG. **4** illustrates forming a scan line on the surface of a diffractive element and imaging the scan line onto the target.
(11) FIG. **5**A illustrates a laser spot scanning across virtual detector pixels on the target.
(12) FIG. **5**B illustrates permissible misalignment while the laser spot is scanning across tall virtual detector pixels on the target.
(13) FIG. **5**C illustrates splitting each of the pixels shown in FIG. **5**B into two sub-pixels.
(14) FIG. **5**D illustrates positioning an array of micro-lenses in front of the detectors, and further illustrates how a relatively small pixel can be collecting light from a relatively large virtual pixel.
(15) FIG. **6**A illustrates the laser of the LADAR system being continuously ON.
(16) FIG. **6**B illustrates the laser being modulated with one short pulse per pixel.
(17) FIG. **6**C illustrates the laser being modulated with multiple short pulses per pixel.
(18) FIG. **7** illustrates synchronizing the clock of an electronic control system with the scanning process, so that there is a constant integer number of clock pulses per scanning cycle.
(19) FIG. **8** illustrates the electronic control system of FIG. **7**.
(20) FIG. **9**A, FIG. **9**B, and FIG. **9**C illustrate a top view, a front view, and a side view of a first embodiment of an improved optical scanning system directed to concentrating more captured light onto one or more of a plurality of pixels.
(21) FIG. **9**D is a perspective view of the embodiment illustrated in FIGS. **9**A-**9**C.
(22) FIG. **9**E is a perspective view illustrating an improved optical scanning system similar to the one shown in FIGS. **9**A-**9**D, but where a Galilean telescoping arrangement is used.
(23) FIG. **9**F illustrates the improved optical scanning system of FIGS. **9**A-**9**D, where a Keplerian telescoping arrangement is used.
(24) FIG. **10**A is an enlarged detail view of one of the detection pixels shown in FIG. **9**D, showing that each pixel may be composed of an array of small sized sub-pixels, and in which the scanning line **22** is parallel to the array of sub-pixels.
(25) FIG. **10**B is view of the detection pixels shown in FIG. **10**A, but where the scanning line **22** is aligned at an acute angle relative to the array of pixels.
(26) FIG. **11**A and FIG. **11**B illustrate a top view and a perspective view of a second embodiment of an improved optical scanning system directed to concentrating more captured light onto one or more of a plurality of pixels.
(27) FIG. **12**A is an enlarged detail view of one or the detection pixels shown in FIG. **11**B, showing that each pixel may be composed of a grid of small sized sub-pixels, where each grid has five rows and seven columns of square-shaped sub-pixels, and in which the scan line is parallel to the sub-pixels.
(28) FIG. **12**B shows the view of the detection pixels shown in FIG. **12**A, in which the scan line is at an acute angle to the sub-pixels.
(29) FIG. **12**C shows the view of the detection pixels shown FIG. **12**B, with the scan line being at an acute angle to the virtual sensors, but where the image spots do not fall within the horizontal center of the sub-pixels.
(30) FIG. **13**A is a color photographic image of a scene with objects at various ranges from the camera;
(31) FIG. **13**B shows colorized LADAR image data with objects in the field color-coded according to range, which image data was obtained using a prototype of one LADAR embodiment described

herein.

DETAILED DESCRIPTION OF THE INVENTION

(32) As used throughout this specification, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than a mandatory sense (i.e., meaning must), as more than one embodiment of the invention may be disclosed herein. Similarly, the words “include”, “including”, and “includes” mean including but not limited to.

(33) The phrases “at least one”, “one or more”, and “and/or” may be open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “one or more of A, B, and C”, and “A, B, and/or C” herein means all of the following possible combinations: A alone; or B alone; or C alone; or A and B together; or A and C together; or B and C together; or A, B and C together.

(34) The word “pixel” is used throughout this specification to denote a discrete element of an optical sensor, such as a CCD, a CMOS device, an APD, etc.

(35) Also, the disclosures of all patents, published patent applications, and non-patent literature cited within this document are incorporated herein in their entirety by reference. However, It is noted that the citing of any reference within this disclosure, i.e., any patents, published patent applications, and non-patent literature, is not an admission regarding a determination as to its availability as prior art with respect to the herein disclosed and claimed apparatus/method.

(36) Furthermore, any reference made throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection therewith is included in at least that one particular embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Therefore, the described features, advantages, and characteristics of any particular aspect of an embodiment disclosed herein may be combined in any suitable manner with any of the other embodiments disclosed herein.

(37) Additionally, any approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative or qualitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified, and may include values that differ from the specified value in accordance with applicable case law. Also, in at least some instances, a numerical difference provided by the approximating language may correspond to the precision of an instrument that may be used for measuring the value. A numerical difference provided by the approximating language may also correspond to a manufacturing tolerance associated with production of the aspect/feature being quantified. Furthermore, a numerical difference provided by the approximating language may also correspond to an overall tolerance for the aspect/feature that may be derived from variations resulting from a stack up (i.e., the sum) of a multiplicity of such individual tolerances.

(38) In accordance with at least one embodiment of the hybrid LADAR system disclosed herein, a fast 1D scanner/imager hybrid can collect one line of high-resolution ToF data within 6.5 microseconds. If needed, that hybrid can also be coupled to a secondary, slow scanning stage, producing raster ToF frames of thousands of lines at high frame rates, with total pixel throughput of tens or even hundreds of megapixels per second. (It is noted that very often, LIDAR systems, as well as vision systems in general, are classified as either “scanning” or “imaging,” implying either a single emitter and/or detector with a scanned FoV, or a multitude of emitters and/or detectors with stationary FOVs (aka “pixels”), whereas the systems disclosed herein are a hybrid between the two, by providing a single scanning emitter and a multitude of stationary detector pixels).

(39) The following description lists several embodiments of the present invention, which are merely exemplary of many variations and permutations of the subject matter disclosed.

(40) Mention of one or more representative features of a given embodiment is likewise exemplary: an embodiment can exist with or without a given feature, and likewise, a given feature can be part

of other embodiments.
(41) A preferable embodiment of a 1D hybrid scanner/imager is illustrated in FIG. **1**. It comprises a 1D array of photo-detectors **1**, placed in a focal point of a receiving optical system **2**, and a scanning mirror **3**, its axis of rotation perpendicular to the direction of the detector array. A laser beam **5** generated by the laser **4** is directed onto the scanning mirror **3** through a collimating lens **6**. The positions of the optical system **2** and the scanner **3** are adjusted in such way that the fan of imaginary rays **7** emanating from individual pixels of the array, and the fan of real laser rays **8** emanating from the scanning mirror **3** lay in the same plane. Respectively, the laser scan line on the target overlaps with the FOV of the detector array.
(42) Preferably, the scanning mirror **3** is a resonant MEMS mirror. Such mirrors, having dimensions of the reflective area of the order of 1 sq. mm, the resonant frequencies of tens of kHz, and the scan amplitude of tens of degrees, are becoming commercially available. While generally this invention would benefit from the fastest rate of scanning, a general tendency in scanning mirrors is that the faster scanning rate typically leads to narrower scan angle and smaller mirror size, which in turn increases the divergence of the scanned laser beam, thus limiting the number and size of the detectors in the array, and the amount of light that can be collected onto the detectors, especially, at longer ranges.
(43) This invention might further benefit from non-mechanical beam scanners (NMBS), that are undergoing development, although their specifications and commercial availability remain unclear. For example, U.S. Pat. No. 8,829,417 to Krill teaches a phase array scanner, and U.S. Pat. No. 9,366,938 to Anderson teaches an electro-optic beam deflector device. NMBS may allow scan rates, or beam cross-sections, exceeding those of mechanical scanners.
(44) A photo-sensor array should preferably consist of high-sensitivity detectors, as the number of photons arriving to each detector, especially from longer ranges, might be exceedingly small. Significant progress has been recently achieved in design and manufacturing of Avalanche Photo Diodes (APD), working in both a linear, and a Geiger mode, which is also known as a Single-Photon Avalanche Detector (SPAD). The former are reported to be able to detect a light pulse consisting of just a few photons, while the latter can actually be triggered by a single photon, and both types would be suitable for embodiments of the present invention. It should be noted, however, that the most advanced detectors are expensive to fabricate, therefore, the present invention that needs only one row of detectors would be very cost-efficient in comparison with flash LADAR, that would typically require a 2D array of the same resolution.
(45) One of the problems typically encountered in ground-based and vehicular LADARs intended to be used in a populated area is the eye safety hazard presented by its powerful lasers. Therefore, a preferable embodiment of this invention would use a laser with longer infra-red (IR) wavelength, exceeding approximately 1400 nm. Such longer wavelengths are not well-absorbed by human eyes and therefore don't pose much danger. Respectively, much more powerful lasers might be used in this spectral range.
(46) Depending on a particular task the hybrid LADAR is optimized for, one of the following configurations of the slow stage may be used for ground surveillance, vehicular navigation and collision avoidance, control of industrial robots, other applications, or alternatively there may not be any slow stage at all. As depicted on FIG. **2**A, the hybrid 1D LADAR of FIG. **1** is attached to an aerial platform **15**, with the scan direction perpendicular to direction of platform motion.
(47) FIG. **2**B illustrates using a scanning mirror **16**, with its scanning direction being perpendicular to the scanning direction of the hybrid. Combined, they constitute a 2D LADAR preferably having comparable scan angles in both directions. Finally, FIG. **2**C illustrates positioning the hybrid of FIG. **1** on a rotational stage **17**, which gives the combined LADAR a 360 degree FOV in a horizontal plane.
(48) If a scanning mirror is used for the slow stage, its area should be sufficiently large to reflect the entire fan of beams emanating from the fast scanner, as well as the entire fan of rays coming to

the optical reception system. However, scanning mirrors with active area of square centimeters, frequencies of tens of Hz, and scan angles of tens of degrees are commercially available. Thus, an exemplary combination of a detector array with 300 pixels, a fast scanner of 30 kHz, and a slow scanner of 30 Hz would provide a point cloud of ˜18M points a second, assuming that both scanners utilize both scan directions, thus enabling real-time 3D vision with the resolution of approximately 1000×300 pixels at 60 frames a second. Rotation at 60 revolutions per second, or linear motion on board a vehicle would also provide comparable point acquisition rates, without the limitations of the reflective area of the scanning mirror.

(49) Preferably, the optical system **2** is configured to form the FOV of each individual pixel **11** of the detector array to match the divergence angle of the laser beam, which is generally possible and usually not difficult to achieve. In an exemplary embodiment, a laser beam of near-infrared (NIR) light of approximately 1 mm in diameter will have a far-field divergence of approximately 1 milliradian (mrad). A photo-detector pixel 20 μm in size will have similar FOV when placed in a focal plane of a lens with 20 mm focal length.

(50) It might also be advantageous to use additional elements in an optical transmission system to shape the laser beam, such as, for example, a cylindrical lens **23**, as depicted in FIG. **3**A. In this case, the laser may still be nearly-collimated in the scanning direction, but may be tightly focused in the orthogonal direction, as illustrated by FIG. **3**B, which shows the view from direction B in FIG. **3**A. Cylindrical lens **23** subsequently re-collimates the beam in the orthogonal direction as illustrated on FIG. **3**C, which shows the view from direction C in FIG. **3**A. The advantage of such an arrangement lies in the possibility to reduce the dimension of the scanning mirror in the tightly-focused laser beam direction, as well as in reduced eye hazard presented by the laser beam expanded in at least one direction.

(51) In another embodiment of the present invention, the laser beam can be tightly focused in both directions and be scanned across a micro optical element **21**, as illustrated on FIG. **4**, forming a thin scan line on its surface. That scan line is subsequently imaged onto the target by the lens **20**. The lens **20** may be identical to the lens **2** in the optical reception system in front of the detector array, in which case the dimensions of the scan line on the surface of a micro optical element **21** should be similar to that of the detector array, thus insuring the equal divergence of both FOVs of the illuminating laser and the detector. Alternatively, the lenses **2** and **20** may have different magnification, and respectively, the scan line may have different dimensions from the detector array, however, the design should provide for a good overlap of both FOVs on the target, as further illustrated on FIG. **5**A, where virtual pixels **31** denote the projections of the detector pixels **11** onto the target through the optical system **2**, while the laser spot **25** scans the same target along the scan line **22**, thus sequentially illuminating pixels **11** with light reflected from the target. As shown on the figure for clarity, at the moment when the laser spot assumes position **25***a*, it illuminates the detector pixel **11***a*. It should be noted that the desired shape of laser spot **25** may be achieved by a variety of optical methods, as illustrated on FIG. **3** and FIG. **4**, or other methods, without limiting the scope of this invention.

(52) Pixels **11** may have different shapes: for example, FIG. **5**B illustrates pixels with their height exceeding the pitch of the array. For example, pixels of the array may be 20 micrometers wide and 60 micrometers toll. The advantage of such arrangement is that it may accommodate slight misalignment between the laser scan line **22** and the line of virtual pixels **31**. Likewise, the laser spot may also be oblong, or have some other desirable shape. It might also be advantageous to split each of the pixels **11** into two or more sub-pixels **11***b*, **11***c*, as depicted on FIG. **5**C. These sub-pixels would generate redundant information, as the scan spot **25** may cross either of virtual pixels **31***b*, or **31***c*, or both of them, at the same time, and the ToA measured by either sub-pixels or both would be treated as one data point. The advantage, however, may come from the fact that smaller sub-pixels would generally have lower noise level, therefore a laser spot of the same power is more likely to generate a response when illuminating a small sub-pixel, than a larger full pixel.

(53) Some types of the high-sensitivity detectors, notably Geiger-mode APDs, are known to need considerable gaps between active pixels to eliminate cross-talk, thus limiting the fill factor of the arrays consisting of such detectors. To alleviate this problem, an array of micro-lenses **12** may be used in front of the detectors **11**, as depicted on FIG. **5**D, with the micro-lens **12***a* specifically illustrating how a relatively small pixel **11***a* can still be collecting light from a relatively large virtual pixel **31***a*.
(54) It is also preferable to match the total extent of the array's FOV with the total scan angle, for example: 512 pixels, 20 um each, placed behind a 20-mm lens will subtend the angle of ˜29°. Respectively, the total scan angle should be the same or slightly greater to utilize every pixel of the array.
(55) In a further embodiment of this invention, as the laser spot moves along the scan line, it is continuously energized, thus producing a time-domain response in the pixels it crosses, as illustrated on FIG. **6**A, where the graph **10**A represents the intensity (I) vs. time (t) response in the pixel **11***a*. The response starts when the laser beam in position **25***b* just touches the virtual pixel **31***a*, and ends when the laser beam moves to position **25***c*, just outside of the virtual pixel **31***a*. In such an embodiment, the laser power, and hence, the sensitivity is maximized, however, the precision of ToF measurement may suffer, due to the ambiguity of the Time of Arrival (ToA) of a relatively long light pulse.
(56) To alleviate this problem, the laser can be electrically-modulated with shorter pulses, as illustrated by FIG. **6**B. For example, the laser is turned on when the laser spot is in the position **26***d*, producing the response **10***d* in the pixel **11***a*. Since the pulse is narrower, its ToA can be determined with greater precision.
(57) One problem with this modulation method might arise if the laser is energized while its spot falls in between two virtual pixels, therefore each of the real pixels is getting only a fraction of the reflected light. Due to some parallax between the laser scanner and the detector array, the precise overlap between the laser spot and virtual pixels is somewhat dependent on the distance to the target, and hence not entirely predictable. To alleviate this problem, the laser may be modulated with more than one pulse per pixel, as illustrated on FIG. **6**B. In this case, at least one of the consecutive modulation pulses emitted with the laser spot in positions **25***f*, **25***g*, **25***h*, would fully overlap with the virtual pixel **31***a*, thus producing full response **10***g* in the pixel **11***a*. Responses from two other pulses **10***f* and **10***h* might be shared with adjacent pixels and hence be lower.
(58) In any LIDAR system dependent on ToF measurements, it is important to accurately measure the ToA of a light pulse on the detector, as well as the time when the light pulse was emitted. In the present invention, additionally, it is important to precisely know the position of the scanned spot, as it determines which detector pixel will be illuminated. To achieve this, it is preferable to synchronize the clock of the electronic control system, denoted by pulses **50** on FIG. **7**, with the scanning process, in such a way that there is a constant integer number of clock pulses per scanning cycle **51**. The vertical axis (S) on FIG. **7** illustrates a scan angle, while the horizontal axis (t) illustrates time. Since many resonant scanners have unique, non-tunable resonant frequency, the system clock frequency may be changed instead to keep a integer number of clocks per period, preferably, by means of a Phase-Lock Loop (PLL) circuit. As long as this fixed relationship between the system clock and the scan angle is maintained, the system clock may be easily used for precise detector read-out timing, as well as for timing of the laser modulation pulses, if such modulation is used. Additionally, some types of detectors require quenching after they received a light pulse, and arming, to be able to receive the next pulse. Keeping pixels armed when they are not expected to be illuminated is undesirable, as it carries the risk of false positive due to internal noise. In most flash LADARs with such detectors, they are armed all at once, right before the illuminating laser pulse is emitted. In the present invention, it is preferable to arm detectors one-by-one, in the order they are illuminated by the scanning laser beam. As illustrated on FIG. **7**, while the scanning beam proceeds from bottom to top, the pixel **11***d* would be armed by the system clock

pulse **50***d*, while pixel **11***e* will be armed considerably later by the system clock pulse **50***e*.
(59) A preferred embodiment of the electronic control system is illustrated by FIG. **8**. The fast scanning mirror **3** is driven by the driver **63** at the frequency defined by the voltage-controlled oscillator (VCO) **67** through clock divider **65**. Being resonant, mirror **3** tends to oscillate at maximum amplitude while driven at its own resonant frequency and in a specific phase with respect to the drive signal. The mirror phase feedback **71** carries the information about the mirror's phase to the phase-locked loop (PLL) **62**, that compares it to the phase of the clock divider and adjusts the VCO to eliminate any phase error, thus maintaining mirror oscillations close to its resonant frequency.
(60) System clock **68** is derived from the same VCO, thus insuring that all other timing blocks function in strict synchronization with the mirror motion. Specifically, the arming circuit **61** provides sequential pixel arming and the laser driver **64** generate laser pulses in specific relationship to the scanned laser spot, as discussed above. Likewise, the same system clock synchronizes the motion of the slow mirror **16** through the driver **66** in a fixed relationship with the motion of the fast mirror, for instance, one cycle of the slow mirror per 1024 cycles of the fast mirror. Read-out circuit **60** and signal processor **69** may use the same clock as well, although they, unlike other elements discussed above, don't have to be strictly synchronized with the mirror motion. They must, however, be fast enough to be able to read and process data from all pixels within one scan cycle of the fast mirror. 3D vision data **70** is generated based on the ToF measurement coming from the detector array **1** and then supplied to users, such as navigation system of autonomous vehicles or security surveillance system.
(61) It should be noted that the present invention offers a considerable advantage in efficiency over other types of LADARs, especially flash LADARs, where short laser pulses are used to illuminate the entire scene at relatively long intervals. The advantage comes from the fact that in the present invention the laser is energized either continuously, or with a fairly high duty cycle: for example, a LADAR of the present invention using 10 ns pulses per pixel and generating 18M data points per second would have a duty cycle of approximately 35%. A flash LADAR using the same 10 ns pulses, and having the same frame rate of 60 fps, would have the laser duty cycle of only 0.00006%. Consequently, to generate the same average power and attain comparable range, the laser of that hypothetical flash LADAR would need instantaneous power almost 6 orders of magnitude greater. While pulsed lasers capable of producing very short powerful pulses do exist, they are known to have lower efficiency, larger size and higher cost than continuous lasers of the same average power. Aside from inability to deliver high average power, pulsed sources are typically less efficient, more complex, bulkier and costlier, than continuous or high-duty sources. The general physical explanation of lower efficiency is in the fact that the emitted power of photonic sources—lasers or LEDs—is typically proportional to the current, while parasitic losses on various ohmic resistances inside those sources are proportional to the square of the current.
(62) Additionally, high-power pulsed laser sources are typically more dangerous in terms of eye safety.
(63) It should be noted, that it is difficult (although not impossible) to place the scanner at the center of the optical system. At any other position, there will be some parallax between the FOVs of the optical transmission and the reception systems, hence the detailed mapping of the pixels onto a scan line will depend on the distance to the target. However, in a practical system such parallax can be kept to a minimum by placing the scanner in close proximity to the optical system.
(64) The present invention as illustrated by the above-discussed embodiments, would provide serious advantages over other types of LADAR.
(65) In a typical imaging LADAR, the light source emits a short pulse which illuminates all pixels at once. Respectively, each pixel receives only a small fraction of the total back-scattered signal. In the proposed hybrid, only one pixel receives all the back-scattered light emitted at a given moment. If we assumed that the illumination power is the same, then the signal strength on each pixel would

be up by a factor comparable to the number of pixels, i.e. hundreds, if not thousands. In practice, pulsed sources generally have higher instantaneous power than continuous ones, but their average power is still considerably lower.
(66) Conversely, in a typical scanning LADAR, at any given moment only a small portion of the photo-detector is receiving any signal, while the rest only generates noise and contributes to unwanted capacitance. The exception is so-called retro-reflective scanners, where a small detector FOV is directed through the same scanning system. However, this approach only works with large, slow scanners, where the mirrors are large enough to provide sufficient optical collection area for back-scattered light. Contrarily, high-speed scanners are usually tiny, just sufficient to fit the beam of the laser, and are usually of the order of 1 mm.
(67) In either case, as illustrated above, a hybrid appears would have a considerable SNR advantage: higher signal than imaging-only, or lower noise than scanning-only device.
(68) It is anticipated that a hybrid LADAR of the present invention will be able to use a regular laser diode as its illumination source—which is by far the cheapest and most efficient source among those suitable for LADARs.
(69) Also, both cost and power consumption are roughly proportional to the total number of sensors/pixels fabricated by a given technology. So, substituting a 2D array by a 1D array is supposed to considerably reduce both cost and power consumption for the detector array, while the cost and power consumption of both fast and slow scan stages may be considerably lower, than that of the array of sensors or the illuminating laser.
(70) The above embodiments, which were originally disclosed in Applicant's patent application Ser. No. 15/432,105, now issued as U.S. Pat. No. 10,571,574, describe LIDAR/LADAR systems wherein a laser beam is scanned by a mirror oscillating about a single axis and directed towards a target, and the light reflected off the target is imaged by a 1D sensor array. The scanner quickly scans the laser beam and therefore the reflected image from the target falls on any given sensor in the array for a very short time. This arrangement eliminates the need for fast pulsing of the lasers thereby drastically simplifying the laser.
(71) In other embodiments, various sensor types can be used (e.g., photo resistors, photo diodes, avalanche photo diodes, phototransistors, etc.), and there are various advantages and disadvantages of each type. A linear array of Avalanche Photo Diodes (APD) may be utilized; however, the state of the technology today would require that approximately 500 photons be received by each pixel to properly trigger a detection event (Note that many photons are needed to "properly trigger a detection event" in order to reliably overcome the internal noise of the sensor in existing linear-mode APD sensors, and estimates suggest that the number of photons needed can be improved by a factor of ˜4, maybe even up to a factor of 10, but probably not down to a single photon.). In a system with an 80 KHz 1D mirror that produces 160 k scan lines per second (each mirror cycle produces two—one going up and one down), and a linear array of sensors each with 512 sensing pixels that produces a data point for each scan line, there are approximately 82 megapixels per second, and each pixel requires 500 photons. This pushes the laser power required to reach a 200-meter detection range to approximately 10 watts, which is achievable for the LIDAR system disclosed herein (being particularly useful for an autonomous vehicle) using, for example, a 10 watt MOPA (master oscillator power amplifier) laser or a continuous wave (CW) fiber laser.
(72) Reduced laser power and/or increased detection range are each desirable features for a Lidar system. One way to achieve this is to increase the sensitivity of the sensor. With a more sensitive sensor, provided the same amount of reflected light hits the sensor, the Lidar will have an increased range. Alternatively, with a more sensitive sensor, to achieve a Lidar system of equivalent range to one with a less sensitive sensor, less laser power needs to be utilized.
(73) Improved sensor sensitivity can be achieved by decreasing the physical area of each pixel of the sensor, which may be done with an Avalanche Photo Detector (APD). This is due to the fact that the sensor's internal noise goes down as the area is reduced. This results in an improvement in

the signal to noise ratio. However, as the pixel area is reduced, from a system standpoint, it becomes difficult to focus all the reflected light from the target onto one particular pixel.

(74) In several embodiments of the present invention described hereinafter, the improved optical scanning systems are directed to arrangements and apparatus for concentrating more captured light onto a plurality of small-area sub-pixels within each sensor.

(75) FIG. **9**A is a top view of a first such embodiment, and is a 2-dimensional view taken along the Y-axis. FIG. **9**A illustrates a laser **4** emitting a circular beam through a cylindrical lens **90** towards a mirror **3** that oscillates about an axis **3**A. The axis **3**A is orthogonal to the scanning direction of the mirror **3**. It is noted that a cylindrical lens, which is often prescribed to correct astigmatism, has a curved surface that may generally focus light into a line rather than a point, and may compress the image in a direction perpendicular to the line (see e.g., Kee, C. S., Hung, L. F., Qiao, Y., Smith, E. L., "Astigmatism in Infant Monkeys Reared with Cylindrical Lenses," Vision Res. 2003; 43: 2721-2739.; and C. J. R. Sheppard, "Cylindrical Lenses-Focusing and Imaging: a Review," Appl. Opt. 52, 538-545 (2013)). However, the cylindrical lens **90** is configured to have no effect on the beam in the scanning direction and expands the beam in the direction orthogonal to the scanning direction, and the laser beam remains collimated or nearly-collimated in that direction after it reflects from the mirror **3** and after it passes through another cylindrical lens **91**, which also doesn't have any effect in the scanning direction. The extent of such nearly-collimated beam in the scanning direction is shown as beam **94** after the mirror and **93** after the cylindrical lens **91**. As noted above, both cylindrical lenses **90** and **91** are configured to have no effect on the beam in the scanning direction and to expand the beam in the direction orthogonal to the scanning direction, as illustrated on FIGS. **9**B and **9**C Since the minimal extent of the beam on a distant target is limited by diffraction, expanding the beam in the direction orthogonal to the scanning direction permits reduction of the extent of the spot on the target in that direction, resulting in the oval-shape spot **92**. The advantage of the oval shape is that it reduces the extent of the beam in a non-scanning direction, thus increasing the power density. Doing so for the scanning direction may also be desirable, but requires increasing the width of the scanning mirror, and while this limitation can be circumvented with the use of a lenslet array (as described below, and shown FIG. **11**), the scan line is no longer continuous, as it breaks into separate unconnected dots.

(76) To expand the beam in the direction orthogonal to the scanning direction, the system may be configured in either a Keplerian telescoping arrangement using the cylindrical lens **90***a* as seen in FIG. **9**F, or a Galilean telescope arrangement using the cylindrical lens **90***b* as seen in FIG. **9**E. Note that in both cases, the extent of the laser beam in the direction orthogonal to the scanning direction may exceed the extent of the mirror in that direction. Note that FIGS. **9**B and **9**C illustrate a Keplerian arrangement. An additional advantage of the Keplerian arrangement is that the extent of the laser beam in the direction orthogonal to the scanning direction on the mirror can be vanishingly small, and respectively, the mirror itself can be small in that direction.

(77) Accordingly, a swept collimated oval beam **93** is moved across a target creating a series of oval spots **92** on the target. That oval spot is imaged onto the sensor(s) by the receiving optical system **2**, where its image also have an oval shape. By way of example the imaged spot size on the sensor(s) can be 55 um (micrometers/microns) wide in scanning direction, but only a few um (e.g., 3-4 um) tall in non-scanning direction.

(78) FIG. **9**D is a perspective view that includes the components of FIG. **9**A and an optical detection system. (Note that the lenses **90** and **91** in the perspective view of FIG. **9**D and in FIGS. **9**A-**9**C are meant to be illustrative of the shape of a cylindrical lenses, and may not be shown proportionately correct to produce the illustrated oval spot). Referring to FIG. **9**D, the oval spot **92** is shown scanned in the direction of scan line **22**. The light scanned onto the target through optical system **2** forms a series of oval laser spots **92** on a plurality of virtual target pixels **31** (shown as individual virtual pixels **31***a*, **31***b*, **31***c*, **31***d*, and **31***e*), which may be received/imaged by the detection sensors **11** on a chip **10**, which may include a plurality of individual sensors **11***a*, **11***b*,

11*c*, **11***d*, and **11***e*. (Note—the optical system **2** may be an imaging objective or imaging lens, and may be a single lens or a series of such lenses that focuses the spot onto the sensors). The system scans along the scan line **22**, thus sequentially illuminating sensors **11** with light reflected from the target. When viewing the oval spot **92** over time as it is continuously scanned along scan line **22**, the oval spot appears as an illuminated line that is a few um thick in the direction across the scan line **22**, i.e. in the non-scanning direction.

(79) FIG. **10**A shows in greater detail the detection sensors **11***a* through **11***e*. In this embodiment, each of the detection sensors **11***a*-**11***e* may be respectively formed of five rectangular shaped pixels, each of which may be transmitted to form an output image **12** that is composed of output image pixels **12***a*, **12***b*, **12***c*, **12***d*, and **12***e* (see FIG. **9**D). For example, detection sensor **11***a* may be formed of the five sub-pixels **11***a***1**, **11***a***2**, **11***a***3**, **11***a***4**, and **11***a***5**. Similarly, detection sensor **11***e* may be formed of the five sub-pixels **11***e***1**, **11***e***2**, **11***e***3**, **11***e***4**, and **11***e***5** (note that the respective five sub-pixels for each of detection sensors **11***b*, **11***c*, and **11***d* that are shown but not explicitly labelled, would also follow the same grid pattern of labelling).

(80) It is noted that the output image **12** may be comprised of the five pixels (**12***a*, **12***b*, **12***c*, **12***d*, and **12***e*), which may correspond to the five pixels of the five sensors **11***a*, **11***b*, **11***c*, **11***d*, and **11***e*, even though the smallest addressable element(s) of each sensor (e.g., sensor **11***a*) is defined in this embodiment as a plurality of sub-pixels (i.e., sub-pixels **11***a***1**, **11***a***2**, **11***a***3**, **11***a***4**, and **11***a***5**), because in one embodiment only the best signal from those five sub-pixels may be used to represent the entire pixel in the image output **12**. In other embodiments several sub-pixels may be used to represent the entire pixel in the output image, and may thus be treated as the whole pixel value.

(81) When the scanned line **22** is perfectly aligned in the center of the virtual pixels **31***a*-**31***e* being imaged on the target by the sensors **11**-**11***e*, as shown in FIG. **9**D, the reflected image of the spot **92** falls on positions **101***a*-**101***e*, and which are shown to be vertically centered in the sensors **11***a*-**11***e*. Positions **101***a*-**101***e* correspond to the position of individual sub-pixels **11***a***3**-**11***e***3**. In this case, only the signal received from individual sub-pixels **11***a***3**-**11***e***3** need to be processed to determine the time-of-flight information. It should be noted that each sub-pixel is smaller than the sensor size, and therefore the sensitivity of each sub-pixels will be significantly improved as compared to a pixel of the size of the entire sensor area. Further, it should be noted, as compared with previous embodiments where the spot from the scanned laser beam is reflected onto the entire sensor, where the optical arrangement compressed all of the photons into the oval spot of the sensor, each of the individual sub-pixels **11***a***3**-**11***e***3** receives approximately the same number of photons which previously reflected onto the entire sensor. Accordingly, receiving the same number of photons on a smaller sub-pixel area results in a more sensitive Lidar system. This would enable the Lidar to have a longer working range and/or operate with lower laser power.

(82) In optomechanical systems, such as the one illustrated in FIGS. **9**A-**9**C and FIG. **9**D and the ones in FIGS. **11**A-**12**C, it is often impractical to provide perfect optical alignment, particularly over expected temperature variations and in the presence of vibrations, and taking into account atmospheric conditions. Therefore, FIG. **10**B illustrates an embodiment wherein scanning line **22** is at an angle relative to the virtual pixels **31***a*-**31***e* shown in FIG. **9**D, and therefore the reflection of the image spot **92** traverses the sensors **11***a*-**11***e* at an angle along scan line **22**. The system can be configured to have one readout channel (timing circuitry for determining time-of-flight) per sub-pixel, and only the individual sub-pixel which receives the maximum signal within each sensor may be connected to the readout channel using, for example, a Field Effect Transistor (FET). For example, referring to sensor **11**C, in this example, only sub-pixel **11***c***3** receives the reflected spot **101***c*, and therefore only sub-pixel **11***c***3** may be connected via an FET to the readout channel associated with sensor **11***c*. However, for sensor **11***d*, the oval spot **101***d* covers both sub-pixels **11***d***3** and **11***d***4**. In one embodiment, whichever sub-pixel receives the larger signal may be connected via a FET to the readout channel associated with pixel **11***d*.

(83) As a still further embodiment, it is possible to connect two or more sub-pixels within a sensor to the readout channel for that sensor. For example, referring to sensor **11***d* in FIG. **10**B, the oval spot falls on both sub-pixels **11***d***3** and **11***d***4**. The output signals of these two sub-pixels may be summed and connected to the readout channel for sensor **11***d*. As a still further embodiment, the outputs of all the sub-pixels within a sensor can be connected to the readout channel for that sensor. Further, it is possible to connect only those sub-pixels receiving signals to the readout channel. In all cases, the connections between the individual sub-pixels and the readout channels can be fixed at manufacturing, or can be dynamically electrically aligned, such electronic alignment can be dynamically updated at modest frequency—say, once per second or minute—based on statistical processing of the returns. As a still further embodiment, it is possible to provide more than one readout channel per sensor, including up to one readout channel per sub-pixel. While these drawing have been shown with five target sub-pixels **31***a*-**31***e* and five sensors **11***a*-**11***e*, the limited number illustrated are to allow the inventions to be easily described herein. As shown by the above example, it is contemplated that the number of virtual pixels **31** and the number of sensors **11** can number in the 100s and even the thousands, (e.g., in one embodiment, a custom chip may be utilized that has 512 sensors, each being a linear mode APD that is 100 um tall (i.e., 100 um across the array direction) and 24 um wide (i.e., 24 um along the array direction), with 6 um gaps between adjacent sensors. FIG. **13**B shows colorized LADAR image data of the scene shown in the photographic image of FIG. **13**A. The LADAR image of FIG. **13**B is obtained from a prototype constructed in accordance with this embodiment (note—to keep the objects in the LADAR image looking close to their natural proportions, the raw image is squeezed into 800(H)×512(V), covering the field-of-view of 45 degrees×26 degrees). In the LADAR image of FIG. **13**B, the ranges to each of the objects in the FOV are color coded according to the color spectrum (ROYGBIV), with various shades of red being close/closer, green being farther away, blue being even further distant, etc. A range color chart may be provided for each LADAR image, depending upon the images obtained, as shown for example on the bottom of FIG. **13**B. The LADAR image is formed line-by-line, where the lines are vertical, with each line being formed while the laser spot reflected from the target traverses all of the 512 sensors of the array. While this is happening, the slow stage turns horizontally by a small angle, so the next vertical line captures a slightly different area of the target. The slow stage may make one full revolution, covering over 8000 vertical scans, and may thus produce an image of 8000(H)×512(V)=˜4M pixels. Since the lines are vertical, the columns have to be horizontal, which is a somewhat unconventional definition, but is nonetheless used to keep the word “line” for the fast vertical scan direction, as the inverse may seem even more unconventional. Also note that the pixels of the 8000(H)×512(V) image, covering a 360 degrees by 30 degrees FOV, will not be exactly square, with each covering a field of approximately 0.045×0.06 degrees.

(84) It is noted that in one example, there may be one individual read-out channel per each sub-pixel, and the signal from each can be directly compared, which may yield the result that certain sub-pixels produce the highest correlated pulses (signals), and that certain sub-pixels produce nothing but noise, and so logically the results from the “good” (signal yielding) sub-pixels could be combined, and the rest of the sub-pixels can be ignored. Depending on the circumstances, a single sub-pixel range measurement may be corrupted and inaccurate, or even non-existent, but under good conditions each sub-pixel will be able to produce an independent, correct range measurement.

(85) In another example, there may be only one read-out channel for all of the sub-pixels of the sensor (e.g., a 5 by 7 sub-pixel array totaling 35 sub-pixels, numbered from 1 to 35). Assuming that initially sub-pixel numbers 12, 13, 19 and 20 are connected to the read-out channel, and after 8000 scans it is determined that 7000 return pulses were collected, and 1000 pulses were missed; then for the next 8000 scans, horizontal sub-pixels numbers 11, 12, 18, and 19 are connected to the read-out channel, and only 500 pulses were missing; then for the next 8000 scans, sub-pixel numbers 10, 11, 17, and 18 are connected to the read-out channel and only 400 pulses were missing; then, for the next 8000 scans, sub-pixel numbers 9, 10, 16, and 17 are connected to the read-out channel, and

600 pulses were missing; a determination can therefore then be made that sub-pixel numbers 10, 11, 17, and 18 provides an optimal configuration for signal reception. A similar search could be done in vertical direction, and it may be repeated periodically, to compensate for possible changes in the system alignment. Although this may be slower, and less reliable than having one read-out channel per each sub-pixel, it likely would be much cheaper and less power-intensive too. It is also possible to have a combination—such as 4 read-out channels for the 35 sub-pixels, etc.

(86) FIG. **11**A is a top view of another embodiment, and which is a 2-dimensional view taken along the Y-axis. FIG. **11**A illustrates a laser **4** emitting a beam, having a circular cross-section, through a focusing lens **100** towards a mirror **3** that oscillates about an axis **3**A. The axis **3**A is orthogonal to the scanning direction of the mirror **3**. The lens **100** is configured to fit the beam **105** on the surface of the mirror **3**. The beam **105** received by the mirror is reflected as beam **108** towards a micro optical element **106**, the micro optical element is not a lens as conventionally known and understood (i.e., not a lens being a simple convex or concave piece of glass or other transparent substance). In one embodiment the micro optical element **106** is constructed as shown in FIG. **11**A and has a linear array of individual focusing lenslets (refractive or diffractive), e.g., lenslets **106***a*, **106***b*, **106***c*, **106***d*, **106***e*, etc. If the micro optical element **106** is illuminated by a point source, it would create as many images of that point source as there are sub-elements (i.e., lenslets) to create multiple point source images. The array may be, for example, a 1×10 array of lenslets. (Note that in one embodiment the number of lenslets is equal to the number of sensors, and in another embodiment the number of lenslets is different than the number of sensors). Each individual lenslet **106***a*-**106***e* is sequentially illuminated by beam **108**, which is thereby shaped and transmitted as a plurality of beamlets (e.g., beamlet **110** from lenslet **106**D) towards a common refractive lens **101** of relatively large diameter. As the laser beam is scanned across each lenslet (e.g., lenslets **106***a*, **106***b*, **106***c*, **106***d*, **106***e*) of the micro optical element **106**, it creates corresponding points of light (i.e., points **102***a*, **102***b*, **102***c*, **102***d*, and **102***e*), and depending upon the dispersion of the light, if the scanned light falls across a pair of adjacent lenslets, it may be divided between those lenslets. Therefore, if the micro optical element **106** has 512 lenslets, then 512 spots will be created, which may be imaged by 512 sensors. The outputs from lens **101** are collimated sub-beams **103** creating a generally circular point **102**. The micro optical element **106**, in addition to having the plurality of lenslets on a front side thereof, also includes is a curved rear surface **106**R, which refracts the scanned beam toward each of the lenslets.

(87) FIG. **11**B is a 3-dimensional drawing containing the components of FIG. **11**A and an optical detection system. (Note that the micro optical element **106** is represented symbolically in the perspective view of FIG. **11**B). The light from a series of virtual pixels **31** (shown as individual virtual pixels **31***a*, **31***b*, **31***c*, **31***d*, and **31***e*) may be received by detection sensors **11** (shown as individual sensors **11***a*, **11***b*, **11***c*, **11***d*, and **11***e*) as a result of the scanned projection onto the target through the optical system **2**. The sub beams **103** emanating from lens **101** form circular spots **102***a*, **102***b*, **102***c*, **102***d*, and **102***e* which sequentially illuminate a sub-portion of the virtual pixels **31***a*-**31***e* and the circular spots **102***a*-**102***e* from the target are received by detection sensors **11***a*-**11***e* respectively. In this embodiment, when viewing the target area over time (longer than the time required for a single scan line **22**), the optical detection system will scan a discrete set of circular points **102***a*-**102***e* upon target areas represented by the virtual pixels **31***a*-**31***e*. By way of example, the size of individual spot **102** received by the sensors **11***a*-**11***e* size can be 5-10 um in diameter, and the spacing between the spots may be much larger that the diameter. In one embodiment, the size of the sub-pixels of each of sensors **11***a*-**11***e* may be similarly sized, having sides being between 5-10 um to accommodate the entirety of the expected spot size, and in another embodiment, the sub-pixels may be square-shaped with each side being one micron in length, and in other embodiments other sub-pixel sizes may be used, with a lower limit on size today being comparable to the wavelength, e.g., about 1.5 um (seemingly being dictated by quantum mechanics and available technology, although theoretically it might be feasible to make even smaller sub-

pixels). Ideally, only one sub-beam would be lit up at any time. However, it is within the scope of this invention to have, at any given moment, more than one sub-beam lit up, but in space they will be well-separated.

(88) It is also noted that an upper limit on the number of the sub-pixels (i.e., a unit that would produce one data point per scan) is generally dictated by the optical resolution of the scanning mirror, and for practical mirror parameters would be on the order of hundreds of sub-pixels, with one embodiment herein utilizing 512 sub-pixels. Also, there is no upper limit on the pixel size. As a theoretical example, a large LIDAR may have, for example, 512 pixels, each 1 mm wide, i.e., a total length of the sensor array being about half of a meter. While such a LIDAR can be hugely sensitive, it would also be hugely expensive. The practical limits on the total size of the array, at least for use in an autonomous vehicle, is roughly on the order of 10-20 mm, making each sub-pixel reasonably limited to a few tens of um, along the direction of the array. In the perpendicular direction, sub-pixels might be somewhat bigger, but generally, there is no reason to make them very tall. Therefore, in one embodiment, a reasonable limit on the number of sub-pixels per sensor would be ˜20 in the direction of the array, and ˜50 in the perpendicular direction, i.e., a 20 by 50 grid totaling ˜1000 sub-pixels.

(89) FIG. **12**A shows in greater detail the virtual pixels **11***a*-**11***e* of FIG. **11**B. In this embodiment, the first sensor **11***a* has five rows of square sub-pixels **11***ay***1**, **11***ay***2**, **11***ay***3**, **11***ay***3**, **11***ay***4**, and **11***y***5** and seven columns of square sensors **11***ax***1** and **11***ax***2**-**11***x***7** for a total of 35 sensors. Note that another n by m grid of sub-pixels may also be used (e.g., n=7 and m=11 for a 7 by 11 grid of 77 sub-pixels). The sub-pixels in sensors **11***b* through **11***e* are identical to those in **11***a* and it is understood that they would follow the same labelling convention, however the labelling is not shown in the drawings. When the scanned line **22** is perfectly aligned in the center of the virtual pixels **31***a*-**31***e* as seen in FIG. **11**B, the reflected image of the spot **102** falls on positions **102***a*-**102***e* that are horizontally and vertically centered in the sensors **11***a*-**11***e*, and are received within the grid of sub-pixels in each of columns **11***ax***4**, **11***bx***4**, **11***cx***4**, **11***dx***4** and **11***ex***4** and row **11***ay***3**. In this case, it may be advantageous that only the signal received from individual sub-pixels upon which the spot **102***a*-**102***e* is received would be connected to a channel readout for each sensor. It should be noted that the use of greater numbers of sub-pixels is advantageous, i.e., where each sub-pixel is smaller than (i.e., 1/35.sup.th the size of) the sensor size, so that the sensitivity of each sensor will be significantly improved as compared to a sensor of the size of the entire pixel area. Further, it should be noted that unlike the previous embodiments described in the Applicant's issued U.S. Pat. No. 10,571,574, where the projected laser reflected onto the entire pixel, due to the optical arrangement which compressed all of the photons into the circular spot, each individual sub-pixel in the array of 35 sub-pixels per sensor of this embodiment receives approximately the same number of photons which previously were reflected onto the entire sensor. Accordingly, receiving the same number of photons on a much smaller area results in a more sensitive Lidar system. This would enable the Lidar to have a longer working range and/or operate with lower laser power.

(90) FIG. **12**B illustrates an embodiment wherein scanning line **22** is aligned at an angle relative to the virtual pixels **31***a*-**31***e* (from FIG. **11**B) and therefore the reflection of the image spots **102***a*-**102***e* fall upon sensors **11***a*-**11***e* at an angle along scan line **22**. In this embodiment, the first sensor **11***a* has five rows of square sub-pixels **11***ay***1**-**11***y***5** and seven columns of square sub-pixels **11***ax***1**-**11***x***7** for a total of 35 sub-pixels. The sub-pixels in sensors **11***b* through **11***e* are identical to those in sensor **11***a* and follow the same labelling convention, however the labelling is not shown in the drawings. The system can be configured to have one readout channel (timing circuitry for determining time-of-flight) per sensor, and only the individual sub-pixel which receives the maximum signal within each sensor is connected to the readout channel using, for example, an FET. For example, referring to sensor **11***c*, in this example, only one sub-pixel (at **11***cx***4**, **11***cy***3**) receives the reflected spot **102***c*, and therefore only sub-pixel **11***c***3** is connected via a FET to the

readout channel associated with sensor **11***c*. Referring to sensor **11***a*, the circular spot **102***a* covers a portion of a first sub-pixel (i.e., the sub-pixel at **11***ax***4**, **11***ay***5**) and a portion of a second sub-pixel (i.e., the sub-pixel at **11***ax***4**, **11***ay***4**). Whichever of those two sub-pixels receives the larger signal may be connected via a FET to the readout channel associated with sensor **11***d*.

(91) As a still further embodiment, it is possible to connect two or more sub-pixels within a sensor to the readout channel for that sensor. For example, referring to sensor **11***a* in FIG. **12**B, the circular spot **102***a* falls on both the sub-pixels at **11***ax***4**, **11***ay***5** and the sub-pixel at **11***ax***4**, **11***ay***4**. The output signals of these two sub-pixels can be summed and connected to the readout channel for pixel **11***a*. As a still further embodiment, the outputs of all the sub-pixels (e.g., all 35 sub-pixels) within a sensor can be connected to the readout channel for that sensor. In all cases, the connections between the individual sub-pixels and the readout channels can be fixed at manufacturing, or can be electrically aligned dynamically, and such electronic alignment can be dynamically updated at a modest frequency—for example, being once per second or minute—based on statistical processing of the returns. As a still further embodiment, it is possible to provide more than one readout channel per sensor, including up to one readout channel per sub-pixel.

(92) FIG. **12**C illustrates an embodiment wherein scanning line **22** is aligned at an angle relative to the virtual pixels **31***a*-**31***e* (from FIG. **11**B), and wherein the image spots do not fall within the horizontal center of the virtual pixels **31***a*-**31***e*. The reflection of the image spots **102***a*-**102***e* fall upon sensors **11***a*-**11***e* at an angle along scan line **22**. In this embodiment, the first sensor **11***a* also has five rows of square sub-pixels **11***ay***1**-**11***y***5** and seven columns of square sub-pixels **11***ax***1**-**11***x***7** for a total of 35 sub-pixels. Sub-pixels in sensors **11***b* through **11***e* are identical to those in **11***a* and follow the same labelling convention, however the labelling is not shown in the drawings. The system can be configured to have one readout channel (timing circuitry for determining time-of-flight) per sensor, and only the individual sub-pixel which receives the maximum signal within each sensor is connected by, for example, an FET, to the readout channel. For example, referring to sensor **11***a*, in this example, only sub-pixel (**11***ax***2**, **11***ay***5**) receives the reflected spot **102***a*, and therefore only that sub-pixel is connected via a FET to the readout channel associated with sensor **11***a*. Referring to sensor **11***c*, the circular spot **102***c* covers four sub-pixels: (**11***ex***2**, **11***cy***4**), (**11***cx***3**, **11***cy***4**), (**11***cx***2**, **11***cy***3**), (**11***cx***3**, **11***cy***3**). Whichever sub-pixel receives the larger signal may be connected via a FET to the readout channel associated with sensor **11***d*.

(93) As a still further embodiment, it is possible to connect two or more sub-pixels within a sensor to the readout channel for that sensor. For example, referring to sensor **11***c* in FIG. **12**B, the circular spot **102***c* falls on four sub-pixels: (**11***ex***2**, **11***cy***4**), (**11***cx***3**, **11***cy***4**), (**11***cx***2**, **11***cy***3**), (**11***cx***3**, **11***cy***3**). The output signals of these four sub-pixels can be summed and connected to the readout channel for sensor **11***a*. As a still further embodiment, the outputs of all the sub-pixels within a sensor can be connected to the readout channel for that sensor. In all cases, the connections between the individual sub-pixels and the readout channels can be fixed at manufacturing, or can be electrically aligned dynamically, such electronic alignment can be dynamically updated at a modest frequency—for example, once per second or minute—based on statistical processing of the returns. As a still further embodiment, it is possible to provide more than one readout channel per sensor, including up to one readout channel per sub-pixel. While the embodiments and drawings show a limed number of sensors and sub-pixels, it is contemplated that the number of sensors and sub-pixels can be substantially increased, and that the size and geometry of the sub-pixels and sensors can be changed. Further, the embodiments of the present illustrate circular or oval beam shapes, it is understood that alternative beam shapes can be utilized.

(94) While illustrative implementations of one or more embodiments of the disclosed apparatus are provided hereinabove, those skilled in the art and having the benefit of the present disclosure will appreciate that further embodiments may be implemented with various changes within the scope of the disclosed apparatus. Other modifications, substitutions, omissions and changes may be made in the design, size, materials used or proportions, operating conditions, assembly sequence, or

arrangement or positioning of elements and members of the exemplary embodiments without departing from the spirit of this invention.
(95) Accordingly, the breadth and scope of the present disclosure should not be limited by any of the above-described example embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A laser radar (LIDAR) system for an autonomous vehicle, said LIDAR system comprising: a laser configured to emit a beam of light at a wavelength; an optical transmission system configured to shape the beam of light emitted by said laser, and to scan said beam along a plurality of transmission light paths toward a target; an optical reception system configured to collect said laser light reflected from the target along a plurality of reception light paths; an electronic control system configured to synchronize said scan of said beam with a respective time-of-arrival measurement; wherein said optical reception system comprises: a one-dimensional sensor array, each said sensor of said one-dimensional sensor array being sensitive to said wavelength of light emitted by said laser, and configured to detect and image said laser light reflected from the target; wherein each said sensor of said one-dimensional sensor array comprises: an n by m grid of sub-pixels, each said sub-pixel configured to receive said laser light reflected from the target; and one readout channel per sensor, each said readout channel comprising: timing circuitry configured to determine the time-of-flight of said beam from said emission from said laser to said reception of said laser light reflected from the target; a field effect transistor (FET); wherein only one of said sub-pixels within each said sensor is connected to said readout channel through said FET at a time, being any one of said grid of sub-pixels determined by said FET to receive a maximum signal; and wherein said electronic control system is configured to perform a time-of-arrival measurement and determine a range of the target.
2. The optical scanning system according to claim 1, wherein said one-dimensional sensor array comprises an array of avalanche photodiodes.
3. The optical scanning system according to claim 2, wherein said laser is from the group of lasers consisting of: a master oscillator power amplifier (MOPA) laser; and a continuous wave (CW) fiber laser.
4. The optical scanning system according to claim 3, wherein said one-dimensional sensor array comprises a linear array of at least one hundred sensors.
5. The optical scanning system according to claim 2, wherein said one-dimensional sensor array comprises a linear array of 512 sensors; wherein said optical transmission system comprises: a mirror said mirror configured to oscillate at about 80 KHz; and wherein said LIDAR is configured to provide a 200-meter detection range utilizing approximately 10 watts of power for said laser.
6. An optical scanning system configured to obtain a range to a target based on a time-of-flight measurement, said system comprising: a laser, said laser configured to emit a substantially cylindrical beam of light along a first optical path; a first lens, said first lens being a cylindrical lens positioned in said optical and configured to expand said cylindrical beam in a first orthogonal direction to said optical path and to have no effect on said beam in a second orthogonal direction to said optical path, to produce an expanding beam comprising an oval-shaped cross-section; a mirror, said mirror configured to oscillate about an axis; said mirror positioned to scan said expanding beam of light along a plurality of optical paths toward a target surface; a second lens; said second lens positioned in said plurality of optical paths, and configured to collimate said scanned expanding beam to produce a collimated oval beam being scanned across the target to create a series of oval spots on the target; a plurality of detection sensors; a third lens, said third lens positioned between the target and said plurality of detection sensors; a grid of sub-pixels for each said sensor, each said sub-pixel configured to receive at least a portion of one or more of said oval

spots of light reflected from the target.
7. The optical scanning system according to claim 6, wherein said second lens is a cylindrical lens.
8. The optical scanning system according to claim 7, wherein each said sub-pixel is square-shaped.
9. The optical scanning system according to claim 8, wherein each said sub-pixel has sides in a range being between five um and ten um.
10. The optical scanning system according to claim 9, wherein each said sub-pixel has sides in a range being between one um and five um.
11. An optical scanning system configured to obtain a range to a target based on a time-of-flight measurement, said system comprising: a laser, said laser configured to emit a substantially cylindrical beam of light along a first optical path; a first lens, said first lens positioned in said optical and configured to focus said beam; a mirror, said mirror configured to oscillate about an axis; said mirror positioned to scan said focused beam of light along a plurality of optical paths toward a target surface; a second lens and a third lens; wherein said second lens positioned in said plurality of optical paths; said second lens comprising a front surface with a plurality of lenslets, each of said plurality of lenslets positioned in line and sequentially illuminated by said focused beam in said plurality of optical paths; said plurality of lenslets configured to individually focus said light of said plurality of optical paths; said second lens configured to collimate said focused light to create a series of spots on the target; a plurality of detection sensors; a third lens, said third lens positioned between the target and said plurality of detection sensors; a grid of sub-pixels for each said sensor, each said sub-pixel configured to receive at least a portion of one or more of said oval spots of light reflected from the target.
12. The optical scanning system according to claim 11, wherein said second lens comprises a concave rear surface.
13. The optical scanning system according to claim 11, wherein said first lens is configured to fit said beam of light onto said mirror.